

CASE 01 · KNIME MACHINE LEARNING STUDY

Predicting Patient No-Shows in Healthcare

Building a machine-learning pipeline in KNIME to forecast which patients will miss their appointments and giving hospitals a sharper tool to schedule, staff, and serve more effectively.

Field: Healthcare Analytics | Tool: KNIME Analytics Platform

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[GitHub](#)

1. PROBLEM STATEMENT

Background

Missed medical appointments — commonly referred to as patient "no-shows" — represent one of the most persistent operational challenges in modern healthcare systems. Globally, no-show rates in outpatient settings typically range between 15% and 30%, depending on the healthcare system, patient demographics, and appointment type. These missed visits create a cascade of inefficiencies: clinical staff are left idle, expensive equipment sits unused, and other patients who genuinely need care are denied timely access.

The financial and operational cost of no-shows is substantial. Beyond direct revenue loss from unfilled appointment slots, no-shows contribute to extended waiting lists, reduced clinical throughput, and poorer patient health outcomes over time. Healthcare providers have traditionally responded by sending blanket reminder messages — but without a reliable way to predict which patients are most likely to miss their appointments, these efforts are inefficient and resource-intensive.

The Dataset

This project analyses a dataset of 110,527 medical appointment records from public hospitals in Vitória, Brazil. The dataset captures a rich combination of patient-level, appointment-level, and communication variables, making it well-suited for exploring the drivers of no-show behaviour.

No-Show	Count	Rate
No (Attended)	88,208	79.8%
Yes (Missed)	22,319	20.2%
Total	110,527	100%

Approximately 1 in 5 patients failed to attend their scheduled appointment — a rate that represents significant lost capacity across the hospital system. Although the majority of patients do attend, the consistent 20.2% no-show rate highlights a structural problem that warrants a data-driven solution.

What This Project Seeks to Solve

This project aims to move beyond simply measuring the no-show problem and toward predicting and preventing it. By identifying the key patient and appointment characteristics that are most strongly associated with missed appointments, the analysis provides a foundation for building a machine-learning model capable of flagging high-risk patients before their appointment date arrives.

Specifically, the project seeks to answer the following research questions:

- What factors most strongly influence whether a patient attends or misses an appointment?
- How does waiting time between the scheduling date and the appointment date affect no-show rates?
- Do certain neighbourhoods exhibit systematically higher no-show rates than others?
- How does age influence appointment attendance?
- Do SMS reminders reduce no-show rates — or is there a selection bias effect at work?

Impact of Missed Appointments

- Staff time, clinical equipment, and facilities go unused during empty appointment slots
- Other patients waiting for care face longer wait times as a result
- Clinicians experience disrupted schedules and reduced productivity
- Downstream scheduling bottlenecks develop across departments
- Revenue is lost from unfilled appointment capacity

Project Goal

To build a predictive machine-learning pipeline in KNIME that identifies patients at risk of missing their appointments — enabling hospitals to intervene earlier, allocate resources more effectively, and ultimately reduce the burden of appointment non-attendance.

2. DATASET OVERVIEW

Source & Context

The dataset was sourced from Kaggle (Medical Appointment No Shows, published by Joniarroba) and contains records from public hospital appointments in Vitória, Brazil. It covers a wide range of patient demographics, medical histories, appointment scheduling information, and communication variables — making it one of the most commonly used datasets for healthcare attendance prediction research.

Features by Category

- Patient: Age, Gender
- Medical conditions: Hypertension, Diabetes, Alcoholism, Handicap
- Appointment: ScheduledDay, AppointmentDay
- Socioeconomic: Scholarship (welfare programme enrolment)
- Communication: SMS_received (0 = no SMS, 1 = SMS sent)
- Target variable: No-show (Yes = missed, No = attended)

Note on the Target Variable

The column name "No-show" is counterintuitively coded: a value of "Yes" means the patient DID miss their appointment, while "No" means they attended. This naming convention is preserved from the original dataset and is important to keep in mind when interpreting model outputs and confusion matrices throughout this report.

Feature Engineering — New Variables Created

Three new variables were engineered to enrich the analysis and improve model performance:

waiting_days

The difference in days between ScheduledDay and AppointmentDay. This required careful time standardisation: ScheduledDay was stored with exact timestamps while AppointmentDay was stored at midnight (00:00:00). Without correcting for this, same-day appointments incorrectly appeared as negative waiting times. The Time Modifier node in KNIME was used to reset both fields to midnight before calculating the difference.

Age_Group

- Child: Age 0–17
- Adult: Age 18–59
- Senior: Age 60 and above

Waiting_Group

- Same Day: 0 days waiting
- Within 7 Days: 1–7 days
- 1 to 4 Weeks: 8–28 days
- Over 1 Month: 29+ days

Class Imbalance

The dataset is notably imbalanced: 79.8% of records are shows and only 20.2% are no-shows. This imbalance is a core methodological challenge — models trained on imbalanced data tend to learn that predicting "show" is almost always safe, which is precisely the failure mode observed in the baseline Decision Tree model. Addressing this through SMOTE is one of the key interventions in the improved Random Forest model.

3. METHODOLOGY

Analytical Approach

The analysis was conducted using the KNIME Analytics Platform through a structured, reproducible workflow pipeline. KNIME was selected for its visual node-based interface, which allows each processing step to be clearly defined, audited, and re-run. The pipeline followed five sequential stages, progressing from raw data ingestion through to final model evaluation.

Stage 1 — Data Cleaning

- Removed PatientID and AppointmentID columns — these are unique identifiers with no predictive value
- Converted string columns to appropriate numeric and date formats

Stage 4 — Baseline Model (Decision Tree)

- Dataset partitioned 50/50 into training and test sets using the X-Partitioner node
- Decision Tree classifier trained using KNIME's tree learner
- Model applied to the held-out test set
- Performance evaluated using the Scorer node

Stage 5 — Improved Model (Random Forest + SMOTE)

- SMOTE applied exclusively to the training partition to address class imbalance
- SMOTE generates synthetic no-show samples by interpolating between existing minority-class examples — not by duplicating them
- The test set was deliberately left untouched with original data only, ensuring unbiased evaluation

- Validated the dataset for missing, null, or logically inconsistent values
- Corrected time inconsistencies between ScheduledDay and AppointmentDay using the Time Modifier node
- Filtered out neighbourhoods with very small sample sizes to ensure statistically reliable comparisons
- Removed extreme or unrealistic values, including ages below 0

Stage 2 — Feature Engineering

- Created waiting_days using the Date & Time Difference node
- Extracted numeric waiting days via the Duration Part Extractor node
- Classified patients into Age_Group categories using a Rule Engine node
- Classified waiting time into Waiting_Group categories using a second Rule Engine node

Stage 3 — Exploratory Data Analysis

- Used GroupBy nodes to aggregate counts by no-show status across key variables
- Used Pivot nodes to reshape data for side-by-side attended vs missed comparison
- Calculated no-show rates using a Math Formula node: $\text{Yes} / (\text{Yes} + \text{No})$
- Produced bar charts for each key variable

- Random Forest classifier trained on the SMOTE-balanced training data
- Model applied to the unchanged test set and evaluated using the same metrics

A Note on SMOTE and Data Leakage

A common mistake when applying SMOTE is to apply it before splitting the data, which causes synthetic samples to "leak" into both the training and test sets. In this analysis, the split was performed first, and SMOTE was then applied only to the training data. This ensures that the test set contains only genuine, real patient records — and that the reported evaluation metrics reflect true out-of-sample predictive performance.

KNIME Workflow Overview

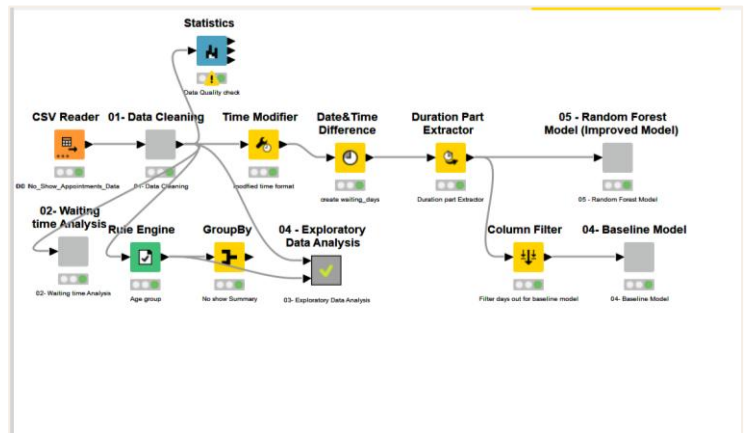


Figure X: KNIME workflow pipeline — CSV import → data cleaning → feature engineering → EDA → dual modelling branches → evaluation.

4. EXPLORATORY DATA ANALYSIS & INSIGHT

Exploratory Data Analysis was conducted to identify patterns and relationships between patient characteristics and appointment attendance. The analysis focused on four key variables — waiting time, neighbourhood, age group, and SMS reminder status — each examined in relation to no-show behaviour. The objective was to understand how these factors influence whether patients attend or miss their scheduled appointments, and to generate insights that would directly inform the predictive modelling phase. Each finding below directly addresses one of the project's research questions.

Figure 1: No-Show Rate by Waiting Time

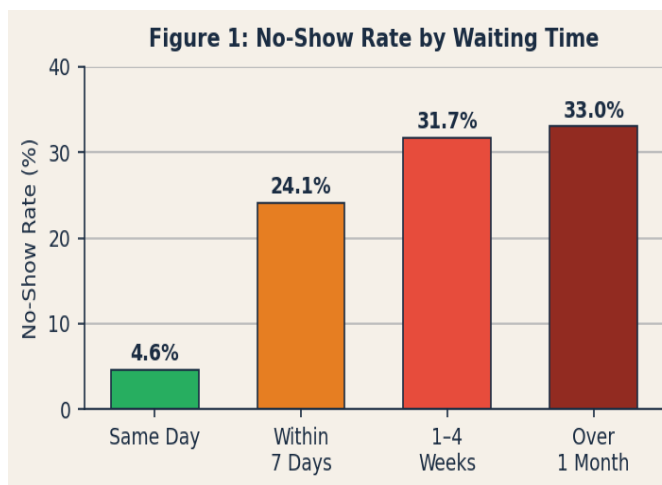


Figure 1: No-show rate rises steeply from 5% (same day) to over 33% (over 1 month).

Figure 2: Top 10 Neighbourhoods by No-Show Rate

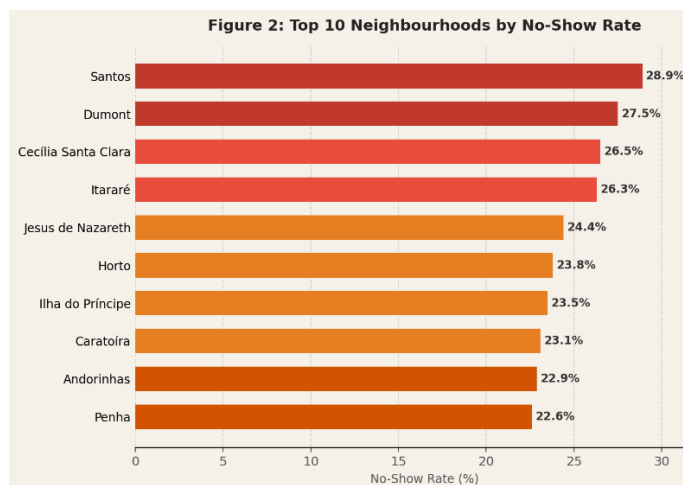


Figure 2: Substantial variation in no-show rates across neighbourhoods. Santos Dumont leads at 28.9%.

The chart reveals a clear and consistent upward trend in no-show rates as waiting time increases. Patients with same-day appointments have a very low no-show rate of approximately 5%, suggesting that when appointments are imminent, patients are highly motivated to attend. As waiting time increases, however, the no-show rate rises sharply — reaching approximately 24% for patients waiting up to seven days, around 32% for those waiting one to four weeks, and over 33% for patients scheduled more than one month in advance.

This pattern likely reflects the way motivation, memory, and life circumstances evolve over time. A patient who books an appointment six weeks in advance has many more opportunities for competing priorities, unexpected events, or simple forgetfulness to intervene. The strong relationship between waiting time and no-show probability makes `waiting_days` the single strongest predictor in this dataset — a finding confirmed by the Random Forest feature importance analysis.

Waiting Group	No-Show Rate	Total
Same Day	~5%	38,563
Within 7 Days	~24%	32,185
1–4 Weeks	~32%	29,396
Over 1 Month	~33%	10,383

Answer to Research Question 2: Yes — waiting time has a strong, direct relationship with no-show probability. Reducing the time between scheduling and appointment is a potentially high-impact operational intervention, though its effectiveness in practice will depend on clinical capacity constraints and patient-level factors not fully captured in this dataset.

Neighbourhood-level analysis reveals substantial variation in no-show rates across different parts of the city. After filtering out areas with very small patient volumes, several neighbourhoods emerge with consistently high no-show rates approaching 25–29%. Santos Dumont records the highest no-show rate at 28.9%, followed by Santa Cecília at 27.5% and Santa Clara at 26.5%.

In contrast, Jardim Camburi — one of the highest-volume neighbourhoods with 7,717 appointments — records a comparatively low no-show rate of approximately 18.9%. This striking contrast between neighbourhoods sharing the same healthcare system strongly suggests that geographic and socioeconomic factors are influencing attendance behaviour. Differences in transport access, local employment conditions, and community health literacy may all be contributing factors.

Neighbourhood	No-Show Rate	Total
Santos Dumont	28.9%	1,276
Santa Cecília	27.5%	448
Santa Clara	26.5%	506
Itararé	26.3%	3,514
Jesus de Nazareth	24.4%	2,853
Jardim Camburi	18.9%	7,717

Answer to Research Question 3: Yes — geography matters significantly. Location-specific interventions such as transport support, community outreach, or satellite scheduling may be warranted in high-risk areas.

Figure 3: No-Show Rate by Age Group

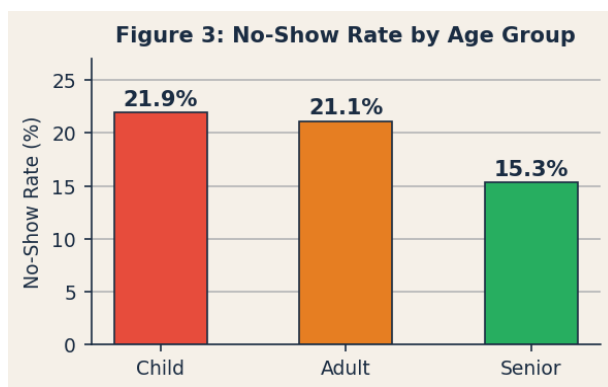


Figure 3: Children (21.9%) and adults (21.1%) have higher no-show rates than seniors (15.3%).

Age group analysis reveals meaningful differences in appointment attendance behaviour across different stages of life. Children exhibit the highest no-show rate of approximately 21.9%, closely followed by adults at 21.1%. Senior patients, by contrast, demonstrate significantly lower no-show rates at 15.3%, making them the most reliable group in the dataset.

The higher no-show rates among children likely reflect the dependency dynamic: a child's attendance is entirely contingent on a guardian's availability, motivation, and schedule. For adults, the comparable rate

Figure 4: No-Show Rate by SMS Reminder Status

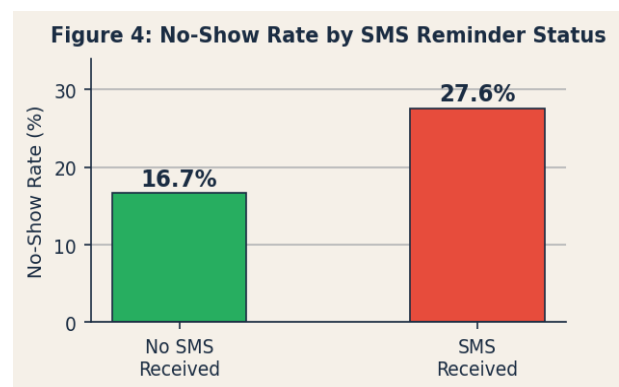


Figure 4: Patients who received SMS show a higher no-show rate — reflecting selection bias, not ineffectiveness.

The SMS analysis produces one of the most counterintuitive findings in the dataset. Patients who received SMS reminders exhibit a higher no-show rate (27.6%) compared to those who received no reminder at all (16.7%). At first glance, this seems to suggest that SMS reminders are actively harmful — but this interpretation would be incorrect.

The most plausible explanation is selection bias. If the hospital's existing system preferentially sends reminders to patients already identified as high-risk — those with long waiting times, prior non-attendance history, or other risk flags — then the SMS group will naturally contain a

may reflect the demands of working-age life: balancing employment, childcare, and other commitments with healthcare attendance. Seniors' lower rates may reflect a stronger perceived necessity for healthcare engagement, or greater availability of time.

Age Group	Attended	No-Show	Rate
Child	21,383	5,997	21.9%
Adult	48,894	13,080	21.1%
Senior	17,931	3,242	15.3%

Answer to Research Question 4: Yes — age is a significant predictor of attendance. Children and adults are the priority groups for intervention strategies targeting younger patients.

disproportionate number of patients who were already likely to miss their appointments. The reminder did not cause the higher no-show rate; rather, the same underlying risk factors that triggered the reminder are also driving the non-attendance.

This means SMS_received should be treated as a risk indicator in the model, not as a causal factor. To properly evaluate whether SMS reminders reduce no-shows, a controlled A/B trial would be needed — randomly assigning reminders to patients regardless of perceived risk, then comparing outcomes between the two groups.

SMS Status	No-Show Rate	Total
No SMS Received	16.7%	75,045
SMS Received	27.6%	35,482

Answer to Research Question 5: The raw data shows SMS is associated with higher no-show rates, but this reflects selection bias — not reminder ineffectiveness. A causal relationship cannot be established from this observational data alone.

5. BASELINE MODEL — DECISION TREE

Model Setup

- Algorithm: Decision Tree classifier (KNIME Decision Tree Learner)
- Dataset split: 50/50 train/test using the X-Partitioner node
- Test set size: 55,264 records
- No class-imbalance correction applied at this stage
- Evaluated using the Scorer node (accuracy, recall, Kappa, confusion matrix)

How the Decision Tree Splits

The Decision Tree learns binary rules by repeatedly splitting the data on the variable that best separates attendees from no-shows at each node. The tree's first and most important split — the root node split — is on `waiting_days`, with the critical threshold at 12 days:

Branch	Show Rate	No-Show Rate	Risk
Waiting ≤ 12 days	95.3%	4.7%	Very Low
Waiting > 12 days	71.5%	28.5%	High

This split confirms the EDA finding: patients scheduled within 12 days are highly reliable, while those with longer waits represent a substantially elevated risk group. The tree then continues splitting on secondary variables to refine predictions further, but no subsequent split approaches the predictive power of the initial `waiting_days` threshold.

Performance Metrics

Metric	Result
Overall Accuracy	77.7%
Recall — Show	93.9%
Recall — No-Show	13.6%
Precision — No-Show	36.0%
F1-Score — No-Show	0.200
Cohen's Kappa	0.098
Test Set Size	55,264

Interpretation

On the surface, 77.7% accuracy appears reasonable. However, this figure is deeply misleading. Because approximately 80% of patients in the dataset attend their appointments, a model that simply predicted "will attend" for every single patient — without any actual learning — would also achieve roughly 80% accuracy. Accuracy alone is therefore an entirely inadequate metric for this problem.

The true failure of the baseline model becomes clear when we look at recall for the no-show class: just 13.6%. This means that for every 100 patients who will genuinely miss their appointment, the Decision Tree only identifies 14 of them — leaving 86 undetected. From a hospital operations perspective, this is near-useless: the model fails to warn staff about the vast majority of upcoming empty slots.

Cohen's Kappa of 0.098 (on a scale where 0 = random guessing and 1 = perfect agreement) confirms that the model's performance barely exceeds chance. The model is heavily biased toward predicting attendance — the majority class — at the expense of any meaningful no-show detection.

Confusion Matrix

	Predicted: Show	Predicted: No-Show
Actual: Show	41,404 ✓	2,700 ✗
Actual: No-Show	9,641 ✗	1,519 ✓

The model correctly identified 41,404 patients who attended (true negatives) and 1,519 who missed (true positives). However, it incorrectly classified 9,641 actual no-shows as likely to attend (false negatives) — the most costly type of error for a hospital trying to anticipate empty slots.

Why This Matters

This baseline model is not suitable for deployment without significant improvement. The failure to detect no-shows — the precise outcome the hospital needs to anticipate — makes it operationally inadequate. This result directly motivates the next step: applying Random Forest with SMOTE to correct for class imbalance and substantially improve minority-class detection.

6. IMPROVED MODEL — RANDOM FOREST WITH SMOTE**Why Random Forest?**

A Random Forest is an ensemble learning method that constructs a large number of individual decision trees during training and combines their predictions by majority vote. This approach has two key advantages over a single Decision Tree. First, by using random subsets of training data and random subsets of features for each tree, the algorithm is far less susceptible to overfitting — each tree sees a slightly different view of the data, and the ensemble averages out individual errors. Second, the forest is better able to capture complex, non-linear interactions between variables, such as the combined effect of long waiting time, young age, and high-risk neighbourhood on no-show probability.

Why SMOTE?

SMOTE — Synthetic Minority Over-sampling Technique — addresses class imbalance by generating new, synthetic examples of the minority class (no-shows) in the training set. Rather than simply duplicating existing minority-class records (which would not add new information), SMOTE creates synthetic samples by interpolating between pairs of real no-show cases in feature space. This gives the model a richer and more varied set of no-show examples to learn from, reducing the bias toward the majority class that crippled the baseline model.

Critical implementation note: SMOTE was applied only to the training partition after the 50/50 split. The test set contains exclusively original, unmodified patient records. This prevents synthetic samples from influencing evaluation metrics and ensures that reported performance figures reflect genuine predictive ability on real-world data.

Model Parameters

- Number of trees: 100
- Feature selection per split: square root of total features
- Train/test split: 50/50 via X-Partitioner
- Test set: 52,925 records (post-SMOTE training set differs in size)
- Evaluated using Scorer node: accuracy, recall, precision, Kappa

Confusion Matrix

	Predicted: Show	Predicted: No-Show
Actual: Show	17,432 ✓	5,934 ✗
Actual: No-Show	9,030 ✗	20,529 ✓

Performance Metrics

Metric	Result
Overall Accuracy	71.7%
Recall — Show	74.6%
Recall — No-Show	69.5%
Precision — No-Show	77.6%
F1-Score — No-Show	0.733
Cohen's Kappa	0.435
Test Set Size	52,925

Interpretation

The Random Forest model represents a dramatic improvement in the outcomes that actually matter. No-show recall has jumped from 13.6% to 69.5% — meaning the model now correctly identifies approximately 7 out of every 10 patients who will miss their appointment. This is the critical figure for hospital operations.

Precision for the no-show class stands at 77.6% — when the model flags a patient as likely to miss, it is correct more than three quarters of the time. The F1-score of 0.733 confirms that the model performs well across both precision and recall dimensions, not just one.

Cohen's Kappa of 0.435 now indicates moderate agreement — a meaningful and substantial improvement over the near-random performance of the baseline model. The slight decrease in overall accuracy (71.7% vs 77.7%) is expected and deliberate: it reflects the model's shift away from defaulting to 'show' predictions, which in turn enables it to catch a far greater proportion of actual no-shows.

7. MODEL COMPARISON

The table below provides a direct, side-by-side comparison of all key performance metrics between the baseline Decision Tree and the improved Random Forest model. Overall accuracy is presented for completeness but should not be used as the primary criterion — the focus is on detecting no-shows, making recall, precision, and F1-score for the no-show class the most meaningful indicators of real-world usefulness.

Metric	Decision Tree (Baseline)	Random Forest + SMOTE	Change
Overall Accuracy	77.7%	71.7%	↓ Expected trade-off
Recall — Show	93.9%	74.6%	↓ Now balanced
Recall — No-Show	13.6%	69.5%	↑ Major gain (+56pp)
Precision — No-Show	36.0%	77.6%	↑ Strong (+42pp)
F1-Score — No-Show	0.200	0.733	↑ Exceptional (+0.533)
Cohen's Kappa	0.098	0.435	↑ Moderate agreement
Test Set Size	55,264	52,925	Different due to SMOTE training expansion

The key improvement is in no-show recall: from 13.6% to 69.5% — a 56 percentage point gain. The slight drop in overall accuracy from 77.7% to 71.7% is a deliberate and expected trade-off, reflecting the model's shift from majority-class bias toward balanced, operationally meaningful predictions.

8. FEATURE IMPORTANCE

How Feature Importance Is Calculated

In a Random Forest, feature importance is measured by tracking how frequently each variable is used to split nodes across all the trees in the forest, and how much each split reduces the model's impurity (how much better the data is separated into shows and no-shows after the split). Variables that appear more frequently in high-impact splits are assigned higher importance scores. The importance scores in this analysis were derived from the KNIME Tree Ensemble Column Statistics node, which aggregates split statistics across the 100 trees in the Random Forest model.

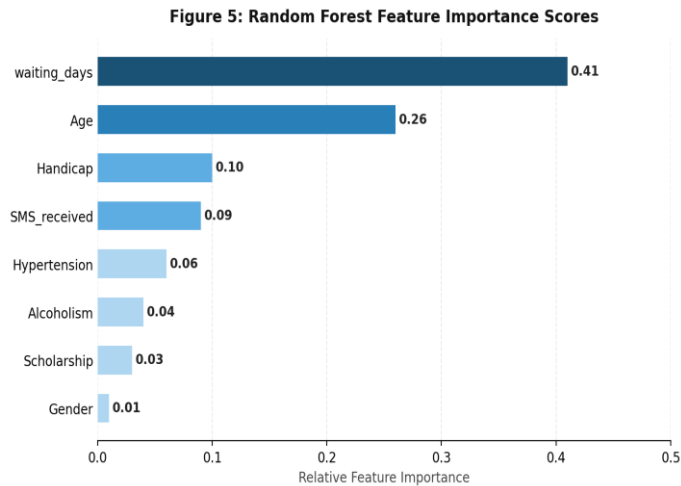


Figure 5: Feature importance scores derived from the Random Forest model.
waiting_days is dominant

Results & Interpretation

The Random Forest model's feature importance rankings are strikingly consistent with the patterns identified during Exploratory Data Analysis, providing strong validation that the model has learned genuine, meaningful relationships rather than statistical noise.

Feature	Importance	Direction
waiting_days	Highest	More days = more risk
Age	High	Younger = more risk
Handicap	Moderate	Contextual
SMS_received	Moderate	Risk indicator
Hypertension	Lower	Mild effect
Alcoholism	Lower	Mild effect
Scholarship	Lower	Mild effect
Gender	Lowest	Negligible effect

waiting_days is the most important feature by a clear margin — confirming that the length of time between booking and appointment is the dominant driver of no-show risk. The model relies heavily on scheduling delay, suggesting that operational inefficiencies in appointment lead time strongly influence attendance behaviour. Patients who must wait more than four weeks face a fundamentally different motivational and logistical context than those seen within days of booking: competing priorities, evolving health perceptions, and

The dominance of waiting_days is consistent with the EDA finding that waiting time is the single strongest driver of no-show behaviour across all patient groups. Its outsized importance score suggests the model relies heavily on scheduling delay as its primary signal — implying that operational inefficiencies in appointment lead time are not merely correlated with non-attendance but are structurally embedded in the patterns the model has learned. This alignment between EDA and model output provides strong validation that the model has captured genuine relationships rather than statistical noise.

the simple erosion of urgency all accumulate over time. Reducing booking lead times, particularly for patients already identified as high-risk, is a potentially useful operational lever, though its feasibility will depend on clinical capacity and departmental scheduling constraints.

Age ranks second, consistent with the EDA finding that children and adults miss appointments at significantly higher rates than seniors. Younger patients should be prioritised in any targeted intervention strategy.

SMS_received ranks fourth, confirming its role as a risk marker rather than a causal factor. The model uses it as a signal that a patient has been identified as high-risk, not as evidence that reminders cause no-shows.

Gender ranks lowest, with negligible predictive value. Gender-based interventions would be unlikely to improve attendance and are not recommended.

9. BUSINESS IMPLICATIONS

The value of a predictive model lies not in the numbers it produces, but in the operational decisions those numbers enable. The following implications outline how the Random Forest model can be translated into concrete, actionable improvements in hospital scheduling, staffing, patient communication, and resource allocation.

1. Quantified Impact on No-Show Detection

The dataset contains 22,319 no-show appointments. With a recall of 69.5%, the model would correctly flag approximately 15,512 of these in advance. If the hospital converts even 30% of flagged high-risk patients into kept appointments through proactive outreach — phone calls, flexible rescheduling, transport assistance — that represents approximately 4,654 recovered appointments per year. This is a conservative estimate; higher conversion rates are achievable with well-designed intervention protocols.

2. Targeted Patient Interventions

- Focus reminder calls and outreach on patients the model flags as high-risk — rather than contacting all 110,000+ patients indiscriminately
- Implement a tiered risk response: patients with $P(\text{No-Show}) > 80\%$ receive a direct phone call; those with $P(\text{No-Show})$ between 50% and 80% receive an SMS and flexible rescheduling offer
- Offer earlier appointment slots to high-risk patients where clinically feasible, reducing the waiting time that drives their elevated risk

3. Scheduling Strategy

- Apply strategic overbooking to time slots with high predicted no-show density — particularly for patients with waiting times exceeding 12 days
- Prioritise reliable patient groups (seniors, same-day bookings, lower-risk neighbourhoods) for critical or time-sensitive appointments
- Reduce idle clinician time and improve daily patient throughput by anticipating empty slots before they occur

4. Resource Allocation

- Match staffing levels and equipment readiness to expected actual attendance rather than booked appointment counts
- Reassign appointment gaps in near real-time when high-risk patients are identified early enough for same-day fill
- Reduce administrative burden from managing no-show consequences reactively

5. Location-Specific Interventions

- Santos Dumont (28.9%), Santa Cecília (27.5%), and Itararé (26.3%) require dedicated strategies — generic reminder systems will not be sufficient
- Potential interventions in high-risk areas: transport voucher schemes, community health worker outreach, local satellite scheduling facilities
- Contrast with Jardim Camburi (18.9%) — lower-risk areas require less intensive intervention, allowing resources to be concentrated where they are needed most

6. Improving Data Quality for Future Models

- Collect patient attendance history — prior no-show behaviour is likely to be the strongest predictor not currently in the dataset
- Record distance from patient home to hospital — transport access is a likely driver of the neighbourhood-level variation observed
- Add socioeconomic indicators to capture the contextual factors behind geographic variation in no-show rates

Deployment Recommendation

This model should function as a decision-support tool, not a fully automated system. Model outputs represent probability scores — clinicians and scheduling staff should use these scores as one input

among several when making decisions about which patients to contact, how to prioritise outreach, and how to structure the day's appointment schedule. Human oversight remains essential.

10. LIMITATIONS

Model Performance Limitations

- Overall accuracy of 71.7% means roughly 1 in 4 predictions is incorrect, scheduling decisions based on model output require human oversight and should not be automated without quality-control mechanisms
- Cohen's Kappa of 0.435 reflects moderate agreement, the model performs meaningfully better than chance, but is not a high-confidence predictor for individual patients
- SMOTE improves class balance during training but introduces synthetic data points, the model's learned boundaries may not perfectly reflect the true distribution of no-show characteristics in the population
- The different test set sizes between the two models (55,264 vs 52,925) arise from the SMOTE process expanding the training set; both test sets contain only real records and are comparable for evaluation purposes
- Precision and recall are in tension: improving recall further (catching more no-shows) would require accepting more false positives — patients incorrectly flagged as high-risk. The current model strikes a reasonable operational balance

Data & Methodological Limitations

- SMS_received shows selection bias and cannot be interpreted as causally influencing no-show rates, a randomised controlled trial would be needed to establish true causal effect
- The dataset lacks patient attendance history, prior no-show behaviour is likely the strongest individual predictor that is currently absent from the model
- No income, transport access, or distance-to-hospital data is available; these are plausible drivers of the neighbourhood-level variation observed but cannot be tested
- The analysis is observational, not experimental. Causal relationships between variables and no-show behaviour cannot be confirmed from this data alone
- Results are specific to the Brazilian public hospital context and may not generalise to other healthcare settings, countries, or appointment types without retraining on local data
- Weather, local events, and seasonal factors are not captured in the dataset but likely contribute to no-show variability in practice. The model may inadvertently learn socioeconomic proxies, variables such as neighbourhood, scholarship enrolment, and waiting time correlate with socioeconomic status. Because the dataset lacks direct income or deprivation measures, the model may use these proxies to approximate socioeconomic risk. This raises a concern: if the model is used to triage outreach resources, patients in already-disadvantaged areas may be disproportionately flagged, potentially reinforcing existing inequalities in healthcare access rather than correcting them. Evaluating the model for disparate impact across demographic groups is an important safeguard before operational deployment. The SMS reminder effect may be more confounded than the selection bias explanation alone suggests — variables such as appointment type, chronic condition severity, and GP referral urgency are absent from the dataset. These unmeasured factors likely influenced both whether a patient received an SMS and whether they attended. Without controlling for them, neither the direction nor the magnitude of the reminder's true effect can be estimated, and the selection bias interpretation — while the most plausible — remains unverified. A randomised controlled trial is the minimum methodological standard required to isolate the causal impact of SMS reminders. Historical patterns embedded in the training data may encode and perpetuate past biases. If certain patient groups — defined by age, neighbourhood, or chronic condition — were systematically less well-served by the hospital in the period covered by the dataset, the model will learn those patterns as predictive signal. When deployed, it may reproduce those inequities by directing outreach away from groups that were already underserved, treating historical under-engagement as an intrinsic patient characteristic rather than a consequence of structural barriers.

Regular fairness audits and disaggregated performance analysis across patient subgroups are recommended as part of any responsible deployment framework.

11. RECOMMENDATIONS

Operational Recommendations

- Prioritise reducing waiting time — keep booking lead times below 12 days wherever clinically possible, particularly for patients with other risk indicators such as young age or high-risk neighbourhood
- Implement a risk-tiered outreach system driven by model predictions: direct phone calls for highest-risk patients, SMS for moderate-risk, standard reminders only for low-risk patients
- Apply strategic overbooking to appointment slots with high predicted no-show concentration
- Develop neighbourhood-specific intervention plans for Santos Dumont, Santa Cecília, and Itararé — these areas require more than reminders alone

Model & Data Recommendations

- Collect and incorporate patient attendance history into the model — a patient who has missed three consecutive appointments is a qualitatively different risk from a first-time booker
- Add distance-to-hospital and socioeconomic indicators to the feature set
- Explore gradient boosting algorithms (XGBoost, LightGBM) as potential alternatives to Random Forest
- Re-train the model quarterly as new appointment data accumulates, to ensure predictions remain calibrated to evolving patient behaviour patterns

Deployment Recommendations

- Integrate the model into the hospital scheduling system to generate patient risk scores automatically at the point of booking
- Display risk scores alongside appointment records for scheduling staff — not as automated decisions, but as decision-support information
- Establish a performance monitoring cycle: review confusion matrix metrics quarterly against actual outcomes to detect model drift
- Ensure all scheduling and clinical staff who interact with model outputs receive training on what the scores mean, their limitations, and how they should inform (not replace) professional judgement

Research Recommendations

- Design a randomised controlled trial to establish whether proactive outreach — phone calls, flexible rescheduling, transport support — for model-flagged high-risk patients materially reduces actual no-show rates
- Conduct qualitative research in high no-show neighbourhoods to understand the specific barriers patients face — structural solutions require understanding the structural problems
- Investigate whether the SMS selection bias can be corrected: if high-risk patients are more reliably identified by the model than by the existing manual targeting process, the model could replace that process

12. CONCLUSION

Summary of Findings

This project successfully developed a complete machine-learning pipeline in KNIME to predict patient appointment no-shows, moving from raw data ingestion through to a deployed, evaluated predictive model. The analysis identified waiting time as the dominant predictor of no-show behaviour: same-day appointments carry a no-show rate of just 5%, compared to over 33% for appointments scheduled more than a month in advance. No-show rates also vary significantly by age group and neighbourhood, and SMS_received was identified as a risk indicator rather than a causal factor — a distinction with important implications for how the hospital designs its intervention strategy.

The baseline Decision Tree model achieved 77.7% overall accuracy but detected only 13.6% of actual no-shows — operationally inadequate for a hospital trying to anticipate empty slots. The improved Random Forest model with SMOTE oversampling raised no-show recall to

Practical Value

Applied in practice, the model enables hospitals to anticipate approximately 70% of missed appointments before they occur. This creates a meaningful window for proactive intervention: targeted outreach to high-risk patients, strategic overbooking of at-risk time slots, and better alignment of staffing levels with expected real attendance. The business case is clear — even a 30% conversion rate on flagged no-shows would recover thousands of appointment slots per year from a dataset of this scale.

The model is not without limitations. With an overall accuracy of 71.7% and a Cohen's Kappa of 0.435, it performs meaningfully above chance but is not a high-confidence individual predictor. It should be used as a decision-support tool — one input among several — rather than as an automated decision-maker. Regular retraining, ongoing performance

69.5%, with a precision of 77.6% and an F1-score of 0.733: a substantial and clinically meaningful improvement that shifts the model from majority-class bias toward balanced, actionable prediction.

monitoring, and human oversight are essential to its responsible deployment.

"While not perfect, the model offers meaningful predictive capability and should be used as a decision-support tool to improve healthcare efficiency and reduce the significant operational cost of patient no-shows."

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